Final Iteration Report

Meal Planner App for the Elderly (MealCare)

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1. Iteration Objectives

The final iteration based on improving usability and enhance user experience for elderly users by following UI/UX standards, introduce user-based access for caregivers and elderly, and finalise core features such as barcode scanning and shopping support. This iteration also focused on refining the system based on previous feedback and ensure stability across devices.

2. Features Implemented

2.1 Caregiver Account and Linking System

A separate caregiver access feature introduced. Caregivers can create own account and link to an elderly user using a UUID. This allows independent access while supporting coordination in assistance.

2.2 Barcode Scanning Integration

Barcode scanning was fully implemented using a browser based scanning library combined with Open Food Facts for barcode lookup. Scanned items are matched against the shopping list, and users receive real-time feedback.

2.3 User Authentication and Password Reset

User authentication is implemented using Supabase Auth, including sign-up, login, and password reset functionality. This enables secure, user-specific data storage and access.

2.4 Terms and Basic Security Handling

Basic terms and conditions is introduce to inform users about system usage and data handling, improving transparency and aligning with security considerations.

2.5 User Interface Improvements for Elderly Users

The user interface was refined to suit elderly users:

- Increased font sizes
- Improved contrast
- Simplified navigation
- Clearer interaction elements

These changes improve accessibility and usability.

2.6 UI Fixes and Stability Improvements

- Resolves critical UI inconsistencies, improving cross-browser compatibility and visual stability.
- Optimized front-end rendering, reducing UI bugs and enhancing the overall user experience.

3. React and JavaScript Concepts Used

The following concepts were applied during this iteration:

- React Hooks: useState, useEffect for managing authentication and UI updates
- Context API: for managing global user and session data
- Async JavaScript: handling API calls (Supabase, Open Food Facts)
- Component Reusability: improving UI consistency
- Conditional Rendering: for authentication states and user roles

4. Implemented Screens

The following figures present screenshots of the implemented application at the end of final Iteration. These images demonstrate the core functional features delivered in this iteration and confirm that a potentially shippable product has been deployed.

Fig.1 this screen shows the user login and account creation interface, including password reset functionality and basic terms access.

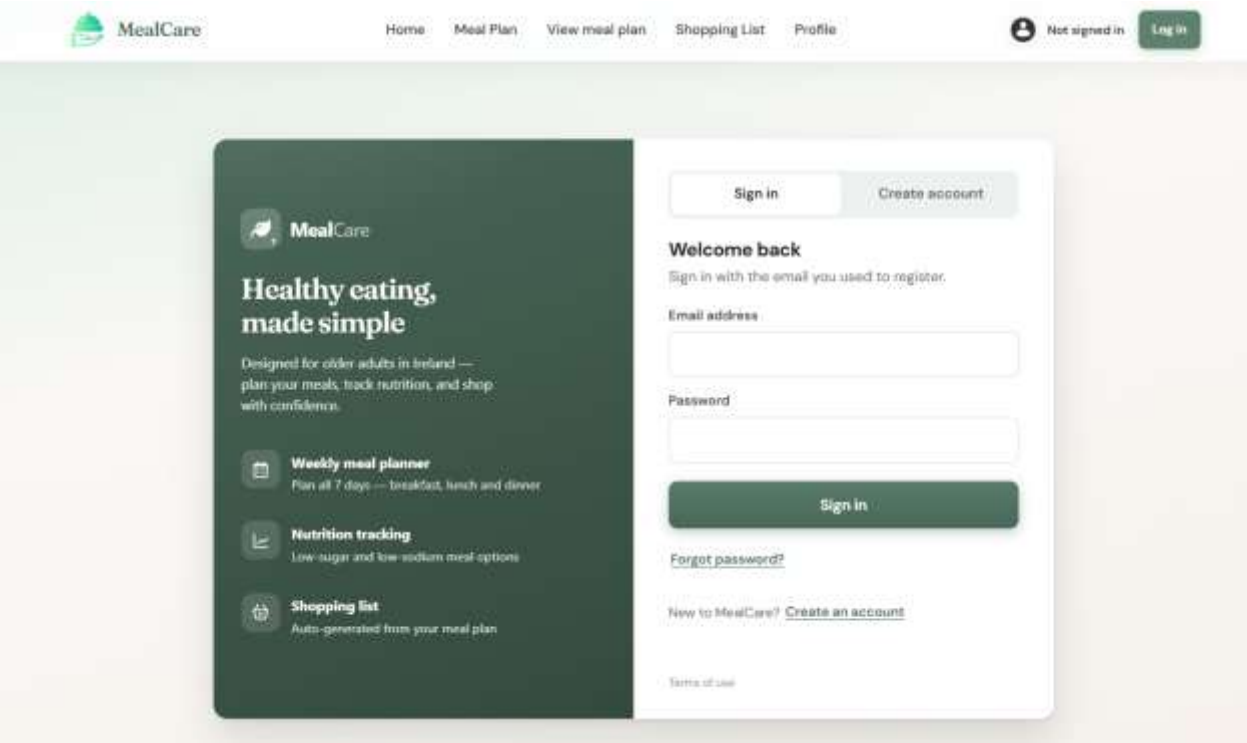

Fig.1 User Authentication Screen (Implemented Version)

Fig.2 this screen demonstrates barcode scanning functionality, including product lookup and matching with the shopping list.

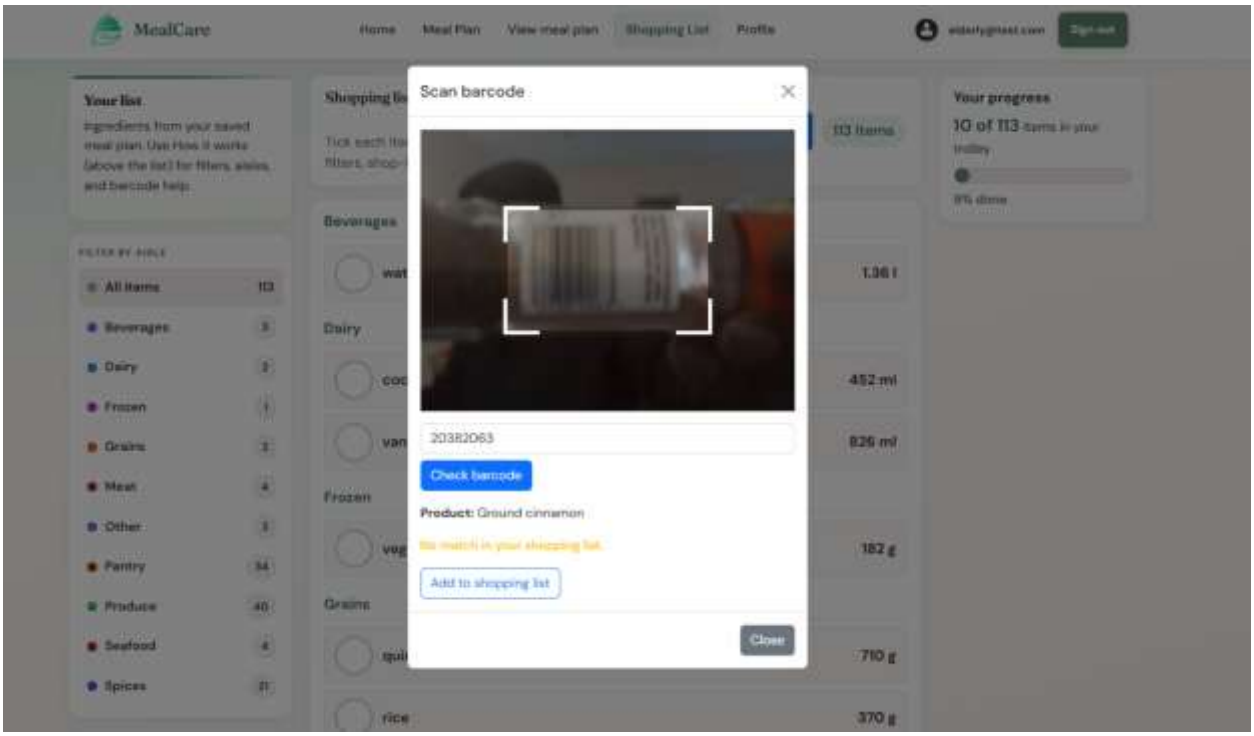

Fig.2 Barcode Scanning Interface (Final Version)

Fig.3 this screen shows the generated shopping list, including item grouping, filtering by category, and progress tracking during shopping.

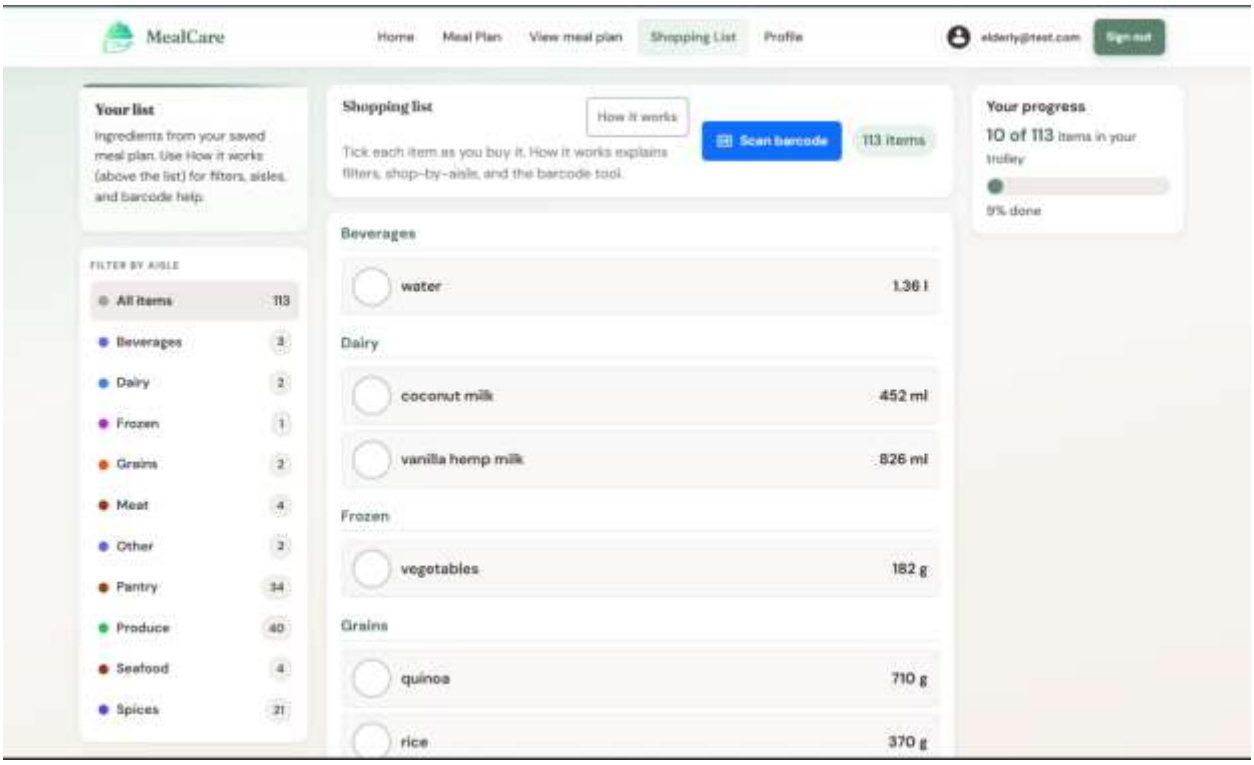

Fig.3 Shopping List Screen (Final Version)

Fig.4 this screen allows users to manage personal details, dietary preferences, health conditions, and budget settings.

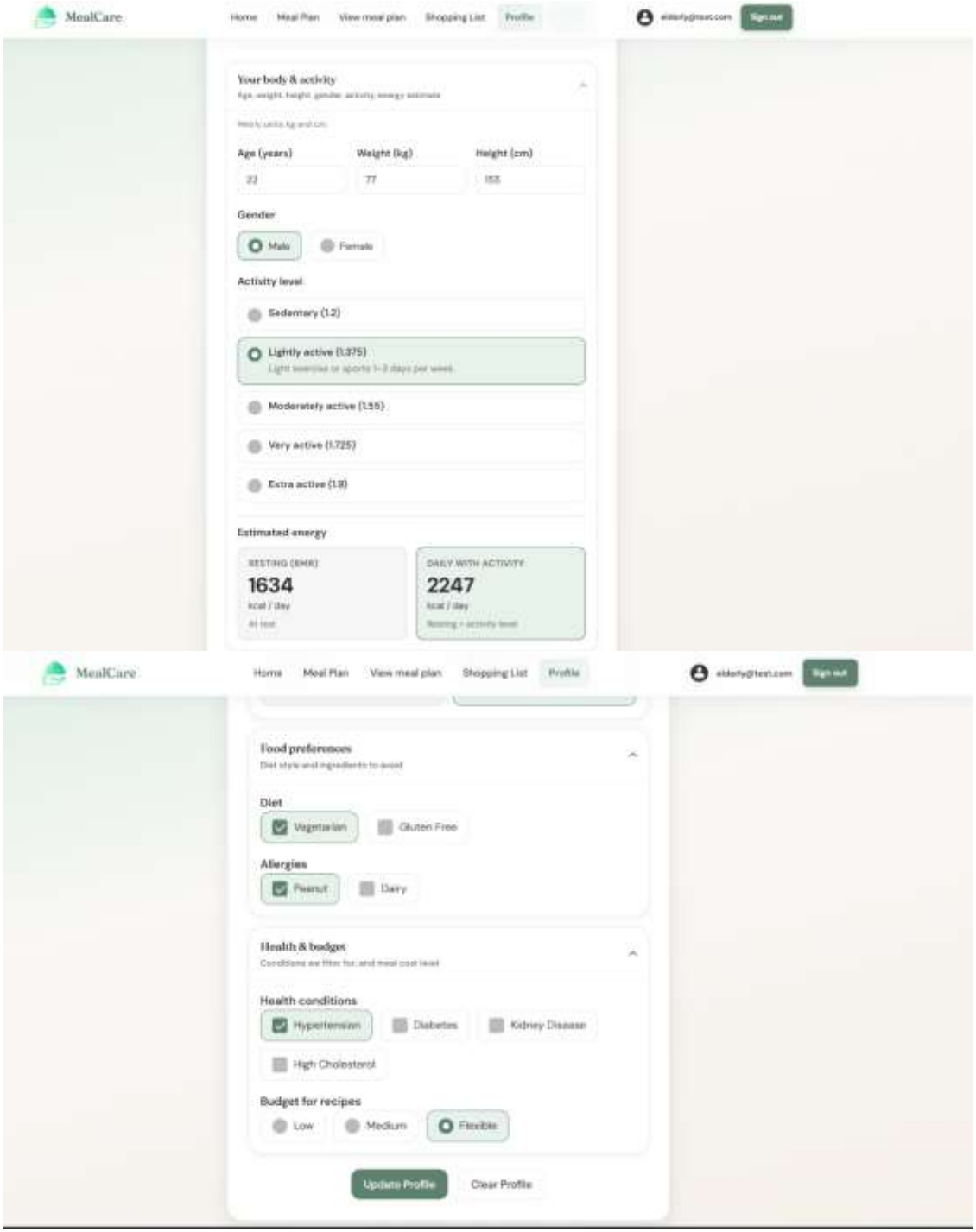

Fig.4 User Profile Management Screen (Final Version)

5. Changes to Vision Document

No major changes were made to the overall vision of the project . The goal of supporting elderly users with meal planning and shopping assistance remained consistent. However, the introduction of caregiver accounts strengthened the coordination aspect originally described.

6. Changes in Research and Design (R&D)

During this iteration, design decisions evolved to support real-world usage:

- Integration of Supabase for authentication and data persistence
- Adoption of external API (Open Food Facts) for barcode data

These changes improved system realism, scalability, and usability.

7. Changes in Specifications

Specifications is refined to match implementation behaviour of newly added features:

The caregiver role access is expanded from using same device/account

The caregiver role expanded from shared usage to a separate account with user linking. Authentication, previously marked as optional, but now implemented to support secure access. Barcode scanning functionality is extended to include real-time product lookup and matching.

8. What Was Not Completed

Some features were not accomplished:

- Integration of grocery delivery service with real vendors.
- Advanced nutrition-based recommendation system
- Multi-user real-time collaboration
- Analytic dashboard for nutrients tracking and analysis.

9. Use of Generative AI

Generative AI tools were used only to assist with content structuring, idea clarification, and summarizing technical details not for direct code generation. In this project, it supported clarity and helped with documentation preparation and code cleaning.

10. Iteration Summary

The final iteration resulted in a fully usable and stable application. Core features such as meal planning, shopping list generation, barcode scanning, and authentication are now integrated into a system.

The application demonstrates strong usability and accessibility for elderly users. The system is ready for deployment for end user.

DECLARATION

I declare that all material in this submission, e.g. thesis/essay/project/assignment, is entirely my/our own work except where duly acknowledged.

I have cited the sources of all quotations, paraphrases, summaries of information, tables, diagrams or other material, including software and other electronic media in which intellectual property rights may reside.

I have cited clearly, exhaustively and adequately in my source code, code generated by Gen AI tools.

I understand that I can only get credit for work that I have done (e.g. understanding other code, interfacing my code with it, modifying it, etc.), but you will not be credited with the third party code itself.

I have provided a complete bibliography of all works and sources used in the preparation of this submission.

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Student Number(s): C00311349

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Date: 23 April 2026

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